MILITARY SPECIFICATION

INDICATOR, ELECTRICAL TACHOMETER, 0-110 PERCENT RPM, 2 INCH SIZE, INTEGRALLY LIGHTED

This specification is approved for use by code 99, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

*1.1 This specification covers one type of hermetically sealed, remote indicating percent speed tachometer indicator, 2 inches in diameter, clamp mounted, with a range of 0-110 percent rpm.

*1.2 Classification. Indicators shall be of the following types, as specified (see 6.2):

ERU-6/A	Covers glass coated with antireflection coating in accordance with MIL-C-675.
ERU-6A/A	Covers glass coated with multilayer anti-reflection coating of high efficiency.

2. APPLICABLE DOCUMENTS

*2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

DD-G-451	Glass, Float Or Plate, Sheet, Figured (Flat, For Glazing, Mirrors And Other Uses)
PPP-B-636	Box, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Hq AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-116	Preservation-packaging, Method Of
MIL-C-675	Coating Of Glass Optical Elements (Antireflection)
MIL-E-5272	Environmental Testing, Aeronautical And Associated Equipment, General Specification For
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification For
MIL-G-9398	Generator, Tachometer, Two-Pole, High Temperature, Aircraft
MIL-G-26611	Generator, Tachometer GEU-7/A Miniature
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification For

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-143	Standards And Specifications, Order Of Precedence For The Selection Of
MIL-STD-794	Parts And Equipment, Procedures For Packaging And Packing Of
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-838	Lubrication Of Military Equipment
MIL-STD-889	Dissimilar Metals
MS24515	Lamps, Incandescent, T-1, Sub-miget Flanged
MS25237	Lamp, Incandescent, Single Contact Midget Flanged Base (T-1-3/4 Bulb)
MS33558	Numerals And Letters, Aircraft Instrument Dial, Standard Form Of
MS33585	Pointers, Dial, Standard Design Of Aircraft Instrument
MS33639	Cases, Instrument, Clamp-Mounted Aircraft
MS33678	Connector, Receptacle, Electric, Integral Mounting.

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 Qualification. The indicators furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.2 and 6.4).

*3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143.

3.2.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.2.2 Standard parts. With the exception in 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 MATERIALS

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.3.2 Metals. Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life.

*3.3.3 Dissimilar metals. Dissimilar metals are defined by MIL-STD-889 and shall not be used in intimate contact with each other unless suitably protected against electrolytic corrosion by means of protective coating or hermetic sealing.

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3.3.4 Fungus-proof materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent or be protected by hermetic sealing.

3.3.5 Corrosive fumes. The materials as installed in the indicator and under the service conditions specified in 3.5 shall not liberate deleterious fumes.

*3.3.6 Protective treatment. When materials are used in the construction of the indicator that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided. Cadmium plating of steel parts must be avoided when satisfactory alternative processes can be used.

3.4 Design and construction. The indicator shall be designed for operation from the varying-frequency alternating-current output of tachometer generators conforming to MIL-G-9398 or MIL-G-26611. The frequency of the generator output shall be proportional to the speed at which the generator is rotating. The design shall be such that with increasing and decreasing speeds the pointers shall move smoothly at a uniform rate consistent with the speed change. The indicator shall be constructed so that no parts will work loose in service. The indicator shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. The indicator shall be constructed so that adjustments and repairs can be easily made by the personnel of operating units and overhaul bases with tools and equipment normally available commercially.

*3.5 Performance. The indicator shall be capable of satisfactory operation when subjected to the following conditions:

a. Indicator starting. Starting and synchronous operation at an indicated speed or 2.5 percent rpm or greater with the following maximum power applied.

(1) Room Temperature - 0.525V at 150 cpm.

(2) -54°C temperature - 0.60V at 150 cpm.

b. Scale error. Scale error operation at room

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temperature, -54°C, -35°C, -5°C, and 71°C temperatures in accordance with tables I and II.

c. Friction error. Friction error not to exceed 1.5 percent at the 5 percent graduation, 0.7 percent at the 20 percent graduation, and 0.5 percent at the 40, 70, 85, and 100 percent graduations.

d. Position error. Position error not to exceed 0.3 percent when positioned 45° each way from the normal operating position with the indicator operating at 100 percent indication.

e. Pointer alignment. When the main pointer indicates 90 and 96 percent, the subdial pointer shall indicate 0 and 6 percent respectively within 0.20 percent.

f. Sealing. Case leakage not to exceed 0.01 micron cubic foot per hour when the filling medium is nitrogen and helium, and 0.1 micron cubic foot per hour when the filling medium is 100 percent helium.

g. Insulation. Withstand the test voltages specified herein for a period of 5 seconds between connector pins A, B, and C and any metal part of the case or connector shell.

h. Magnetic effect. Subject the indicator, operating at 90 percent rpm and nonoperating, rotated in increments of 45° from normal, and positioned 5-½ inches from the center of a free-magnet compass having a field intensity of 0.17 to 0.19 oersted, without deflecting the compass more than 3°.

i. Vibration. Subjected to circular vibration in a plane inclined 45° to the horizontal under the following conditions:

(1) Error. Shall not exceed 0.7 percent rpm when operating at a point between 60 and 90 percent rpm and at an amplitude of 0.009 to 0.011 inches while the frequency is varied from 5 to 50 cps.

(2) Failure. No evidence of failure when operated at 50 percent rpm and vibrated for 3 hours at an amplitude of 0.018 to 0.020 inches while the frequency is cycled between 5 and 50 cps and return once each hour during this test.

j. Indicator current. The current drawn by each of the 3-phase windings shall not exceed 0.3 amp when connected to 21V 4200 cpm supply.

k. Temperature. Subjected to temperature ranging from -62°C to +85°C without failure.

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TABLE I

Calibration Scale Errors

Generator Drive Shaft Speed	Correct Indicated Speed	Scale Error Tolerance	Scale Error Tolerance			
		Room Temperature	-35°C	-54°C	±71°C	-5°C
RPM	PERCENT RPM	+PERCENT RPM		+PERCENT RPM		
0*	0	0.5	0.6	0.6	0.5	0.6
210*	5	0.5	0.6	0.6	0.5	0.6
420*	10	0.5	0.6	0.6	0.5	0.6
630	15	0.5	---	---	---	---
840*	20	0.5	0.6	0.7	0.5	0.6
1260	30	0.6	---	---	---	---
1680*	40	0.8	1.1	1.2	1.0	1.1
2100	50	0.8	---	---	---	---
2520*	60	0.8	1.2	1.4	1.0	1.1
2940	70	0.8	---	---	---	---
3150*	75	0.5	1.0	1.3	0.8	0.8
3360*	80	0.5	1.0	1.4	0.8	0.8
3570*	85	0.5	1.0	1.4	0.8	0.9
3780*	90	0.5	1.1	1.5	0.8	0.9
3990*	95	0.5	1.1	1.5	0.8	0.9
4200	100	0.5	1.1	1.5	0.8	0.9
4410*	105	0.5	1.2	1.5	0.8	0.9
4620	110	0.5	---	---	---	---

NOTE: * Low and High Temperature Test Points.

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TABLE II

Calibration Scale Errors

Generator Drive Shaft Speed	Correct Indicated Speed	Scale Error Tolerance	Scale Error Tolerance			
		Room Temperature	-35°C	-54°C	+71°C	-5°C
RPM	PERCENT RPM	+PERCENT RPM	+PERCENT RPM			
0	0	0.5	0.6	0.6	0.5	0.6
200*	4.76	0.5	0.6	0.6	0.5	0.6
400*	9.52	0.5	0.6	0.6	0.5	0.6
600	14.29	0.5	---	---	---	---
800*	19.05	0.5	0.6	0.7	0.5	0.6
1200	28.57	0.6	---	---	---	---
1600*	38.10	0.8	1.1	1.2	1.0	1.1
2000	47.62	0.8	---	---	---	---
2400*	57.14	0.8	1.2	1.4	1.0	1.1
2800	66.67	0.8	---	---	---	---
3200*	76.19	0.5	1.0	1.3	0.8	0.8
3400*	80.95	0.5	1.0	1.4	0.8	0.8
3600*	85.71	0.5	1.0	1.4	0.8	0.9
3800*	90.48	0.5	1.1	1.5	0.8	0.9
4000*	95.24	0.5	1.1	1.5	0.8	0.9
4200*	100.00	0.5	1.1	1.5	0.8	0.9
4400*	104.76	0.5	1.2	1.5	0.8	0.9
4600	109.52	0.5	---	---	---	---

NOTE: * Low and High Temperature Test Points.

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1. Overspeed and reverse operation. A generator-shaft speed of 5,000 rpm for 5 minutes, and then operated in reverse direction at a generator-shaft speed of 2,500 rpm for 1 minute. The change in indicated speed shall not exceed 0.5 percent.

m. Endurance. Subjected to 1,000 hours operation at 90 ± 5 percent indication without damage.

n. Thermal shock. Subjected to 4 cycles of exposure to water baths maintained at a temperature of +85` ± 2`C and 5` ± 2`C without evidence of leakage when subjected to the sealing test.

o. Precipitation or fog. Subjected for a period of 2 hours at an ambient temperature of 80`C to 90`C, and then immediately immersed in a liquid bath maintained at 0` to 5`C, without evidence of moisture or fog deposit on the inner face of the cover glass.

p.. Salt spray. Subjected to salt spray for a period of 50 hours in accordance with MIL-E-5272 without evidence of external corrosion or deterioration affecting subsequent operation.

q. Lighting tests. Subjected to the lighting tests specified herein.

r. Indicator acceleration. Withstand acceleration from 0 to 30 ±5 percent within a period of 4 seconds.

s. Response. Accelerate from 50 to ±1.5 percent to 100 ±1.5 percent in 2 seconds or less.

t. Internal examination. Subjected to examination internally for corrosion, deterioration, or damage resulting from all tests specified herein.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and drawing number requirements of DOD-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.7 Case. The case shall be in accordance with figure 1 of this specification and shall provide a hermetically sealed enclosure for the internal mechanism of the indicator. The design of the case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of

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any atmosphere to which the indicator may be subjected. The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable and shall not exceed 1/32 inch.

3.7.1 Material. The hermetically sealed case shall be constructed of a suitable material applicable to hermetic sealing, and the material shall be uniform in texture with a smooth external surface.

3.7.2 Finish. The portion of the indicator case exposed to view from the front of the mounting panel shall be finished in durable dull black. The finish applied to the remainder of the case shall be optional, provided that the finish used will in no way prevent compliance with the requirements of this specification.

3.7.3 Filling medium. Where a hermetically sealed case is used the filling medium shall be of at least 98 percent purity, free of dust particles, and contain not more than 0.006 milligram of water vapor per liter (dew point -65°C) at the filling pressure. The filling medium shall be either 100 percent helium or a mixture of 88 percent to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere. If practicable the 100 percent helium filling medium shall be utilized.

3.7.4 Integral lighting. The indicator shall be integrally lighted in accordance with MIL-I-27160.

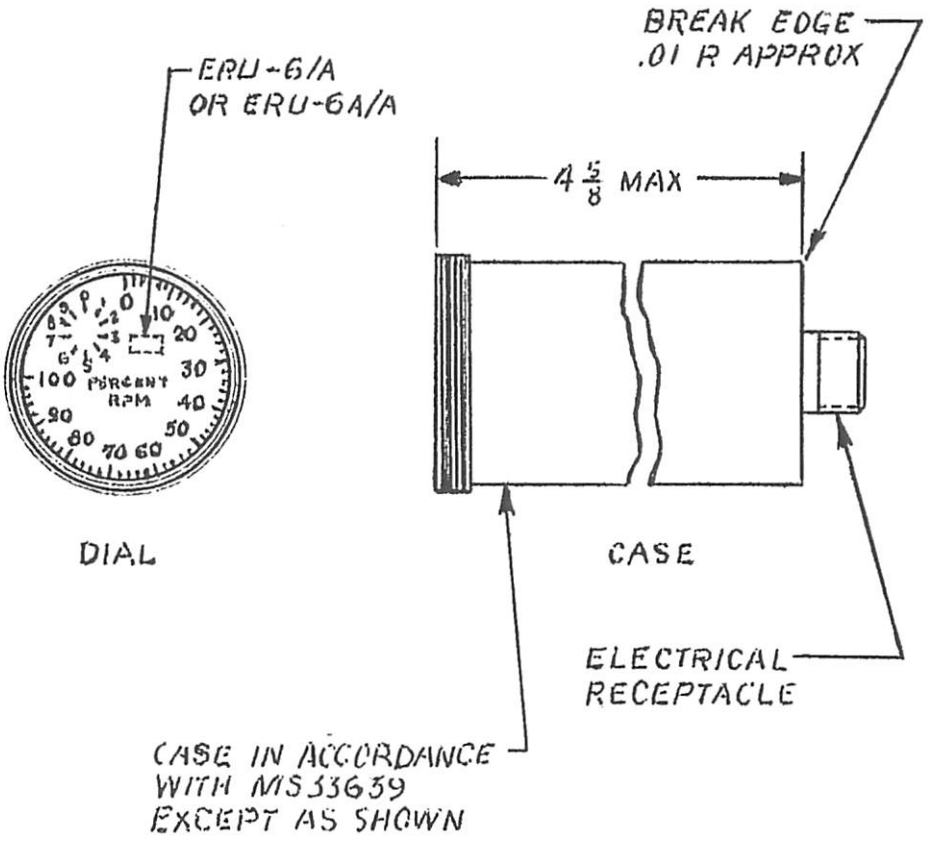
3.7.4.1 Lamp. The light source for the indicator shall be three lamps conforming to either MS25237 or MS24515.

3.7.4.2 Lamp location. The lamps shall be contained within the hermetic seal. The lighting system shall be designed so that, when lamp replacement is necessary, the lamps may be accurately positioned in place. No provision shall be made for lamp replacement in the field. The lamps shall be so installed that easy replacement can be accomplished at a repair depot.

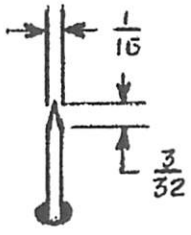
3.8 Cover glass. The cover glass shall be in accordance with DD-G-451, type II quality AA. The thickness of the glass shall be the minimum practicable and entirely suitable for the purpose. Any flaws allowed by the Federal specification shall not interfere with the readability of the instrument.

3.8.1 Dial to cover glass distance. The distance between the inside surface of the cover glass and the indicator dial shall be the minimum practical and shall not exceed 0.188 inch.

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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCES: FRACTIONS ±1/64



SUB DIAL
POINTER

FIGURE 1.

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*3.8.2 Reflecting surfaces. All visible reflecting transparent surfaces shall be provided with a reflection-reducing coating which meets the requirements of MIL-C-675 in addition to withstanding the environmental requirements specified herein. For the ERU-6A/A indicator, the following tolerances shall apply:

Angle of Incidence	Wavelength Millimicrons	Percent Reflectance
0`	460-675	0.6 maximum
0`	425-700	0.5 average
30`	450-625	1.0 maximum
30`	425-700	0.5 average

*3.9 Dial. The dial shall consist of either a flat surface with a recess for the subdial pointer or one flat surface on a single plane. If a recess is utilized, the forward surface of the subdial pointer must lie in the same plane as the main dials. The dial shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated. If screws are used for attaching the dial, they shall be located so that they shall not interfere with any dial marking or prevent proper location of dial marking. The dial background and the shaded protion of the pointers shall be black and shall conform to color No. 37038 of FED-STD-595. The dial shall have a minimum diameter of 1-3/4 inches when measured across the outside ends of the graduations. The diameter of the subdial, when measured across the outside ends of the graduations, shall be 7/16 ± 1/64 inch.

3.9.1 Markings. The dial markings shall conform to figure 1 and shall be durable to withstand usage in service. The 100 percent rpm graduation shall be located on the horizontal center line at the left side of the dial. The dial markings shall be white and the color shall conform to color Nr 37875 of FED-STD-595. Unless otherwise specified, the style and proportions of numerals and letters placed on the dial shall conform to MS33558. Numerals shall distinctly indicate the graduation to which each applies. Any confusion resulting in doubt as to which graduation the numeral applies shall be cause for rejection. The dial graduations shall be uniformly spaced. The dimensions of the markings shall be as follows:

Marking	Height or Length +0.016	Width of Line or Graduation +0.005	Color FED-STD-595
Numerals 0, 10, 20, 30, 40, 50, 60, 70, 80, 90, and 100	0.141	0.020	

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Marking	Height or Length ±0.016	Width of Line or Graduation ±0.005	Color FED-STD-595
Numerals 1, 2, 3, 4, 5, 6, 7, 8, 9, 0	0.094	0.016	
10 percent graduations	0.141	0.025	
2 percent graduations	0.093	0.016	No. 37875
Subdial graduations	0.063	0.010	
Lettering "Percent"	0.093	0.016	
Lettering "RPM"	0.125	0.016	
Shaded Portion of Pointers	----	----	No. 37038
Lettering ERU-6/A	0.063	0.010	No. 37038

3.10 Electrical receptacle. Each indicator shall be provided with a five-pin electrical receptacle conforming to MS33678 located on the vertical and horizontal center lines of the indicator with the polarizing key of the receptacle at the top. The receptacle shall be of the hermetically sealed type. The receptacle shall be provided with a cap or cover and gasket to prevent the penetration of water vapors.

*3.11 Pointers. The lighted area of the subdial pointer shall conform to figure 1. The length shall be such that the outer ends of the pointer will approach, as close as practicable, the inner ends of the subdial calibration marks. The dimensions of the main dial pointer, as viewed from the front of the indicator, shall conform to MS33585-8. The cross section of the pointers shall be suitable for integral lighting and sufficiently rigid to prevent failure due to vibration.

3.12 Indication. The tachometer indicator shall indicate 100 percent speed of the engine when the generator is driven at 4200 rpm. The indicator shall indicate 0 to 100 percent speed in approximately 270° pointer travel. The subpointer shall make one complete revolution for each 10 percent change in indication of the main pointer. It shall be possible to read 110 percent by using the subpointer for the last 10 percent, and the main pointer shall indicate at a point equal to 110 percent rpm when the subdial is indicating 110 percent rpm.

3.13 Wiring. The three leads from the indicator windings shall be designated at B, A, and C (conforming to phases 1, 2, and 3). The leads shall be connected to the B, A, and C pins of the electrical receptacle in such a manner that when 3 phase power with phase rotation sequence B-A-C is applied, the resultant indicator reading is positive. The remaining pins, C and D, shall be used for integral lighting.

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3.13.1 Electrical connections. Electrical connections for counterclockwise generator drive shaft rotation (when viewed from the shaft end), for operating one or two indicators, shall conform to the wiring diagram shown on figure 2. For clockwise rotation reverse leads A and B at generator plug. The 5V lighting wiring shall be as shown on figure 2.

3.14 Stops. The indicator mechanism shall be provided with a stop, which upon reverse operation of the indicator motor shall stop the pointer motion after traveling through an arc equal to not less than 2 percent.

*3.15 Lubrication. Lubrication shall be accomplished in accordance with MIL-STD-838, except that the selection of lubricants used shall be approved by the procuring activity.

*3.16 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.17 Weight. The weight of each indicator shall not exceed 1 pound.

3.18 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.19 Workmanship.

3.19.1 General. The indicator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.19.2 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.19.3 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further, without damage or injury to the screw, bolt or threads.

3.19.4 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

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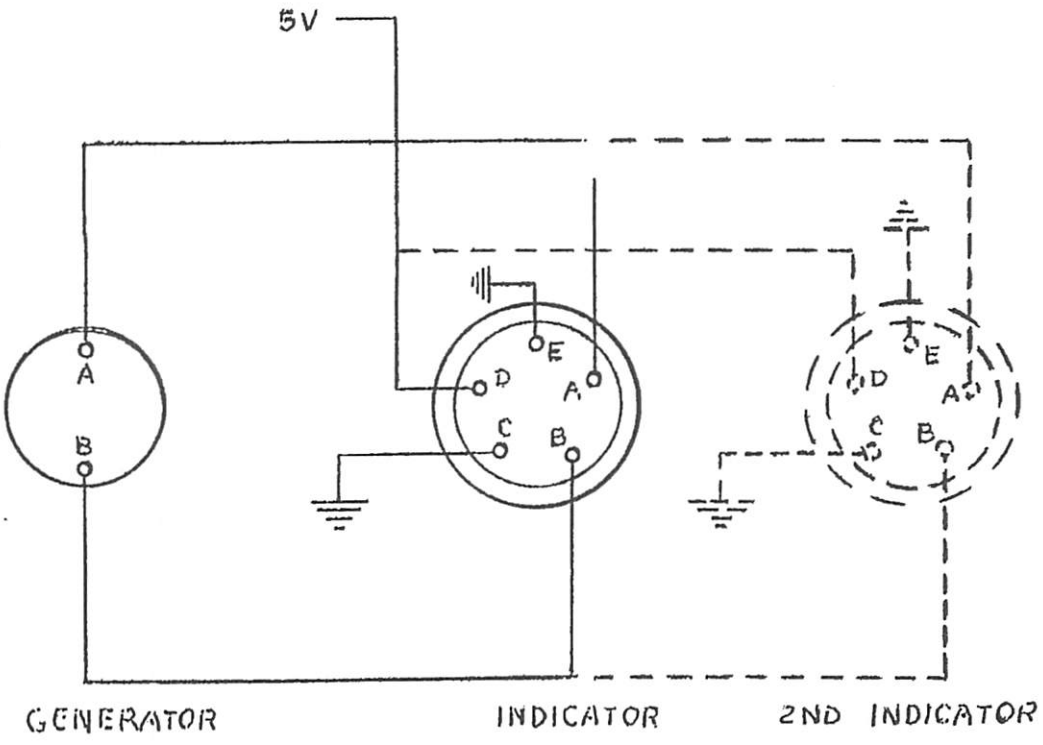


FIGURE 2 Wiring diagram.

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3.19.5 Cleaning. The indicator shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Classification of tests. The inspection and testing of the indicator shall be classified as follows:

a. Qualification tests. - (See 4.2)
b. Acceptance tests. - (See 4.3)

*4.2 Qualification testing.

*4.2.1 Qualification test sample tested by the contractor. The contractor shall subject three indicators to the qualification test specified in 4.2.4.

*4.2.2 Qualification test report. After the contractor completes the qualification tests, he shall prepare a qualification test report according to MIL-STD-831 and furnish three complete copies of the report to the procuring activity.

*4.2.3 Qualification test sample for the procuring activity. Along with the qualification test report, the contractor shall submit sample(s) to the procuring activity who will use it (them):

a. For a review of the mechanical construction of the product.

b. To perform any tests, included in the specification, after reviewing the contractor's test report.

*4.2.4 Qualification tests. The qualification tests shall consist of all the tests specified herein.

4.3 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling plans and tests

4.3.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.5 Test methods:

a. Examination of product

b. Room temperature starting

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c. Indicator acceleration

d. Scale error at room temperature

e. Friction error

f. Position error

g. Pointer alignment

h. Sealing

i. Precipitation or fog

j. Individual tests of MIL-L-27160

4.3.2 Sampling plans and tests.

4.3.2.1 Sampling plan A. One indicator selected at random from each 100 or less produced on the contract or order shall be subjected to the following tests as described under 4.5 Test methods:

a. Individual tests

b. Low temperature starting

c. Low temperature warmup

d. Scale error at -54°C

e. Scale error at -35°C

f. Scale error at -5°C

g. Scale error at high temperature

h. Insulation

i. Magnetic effect

j. Vibration error

k. Sampling plan A tests of MIL-I-27160

*4.3.2.2 Sampling plan B. Unless otherwise specified, 3 indicators selected at random from the first 15 items of the contract or order shall be subjected to the following tests as described under 4.5, Test methods:

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a. Sampling plan A tests

b. Indicator current

c. Low temperature exposure

d. High temperature exposure

e. Vibration failure

f. Overspeed and reverse operation

g. Indicator endurance

h. Thermal endurance

i. Salt spray

j. Sampling plan B tests of MIL-L-27160

k. Response

l. Internal examination

4.3.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.3.2.3.1 Individual tests may continue. For operational and production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.3.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Test conditions.

4.4.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and

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at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified conditions.

4.4.2 Tapping. Except where otherwise specified, the indicator shall be gently tapped or vibrated before a test reading is taken.

4.4.3 Vibration. Unless otherwise specified, the indicator shall be subjected to circular vibration in a plane inclined 45° to the horizontal plane. The vibration shall be within a frequency range of 5 to 50 cps and at amplitudes within 0.009 to 0.020 inch.

4.4.4 Attitude. Unless otherwise specified, the indicator shall be tested in its normal operating position (dial surface shall be held in vertical plane).

4.4.5 Test voltage. Unless otherwise specified, voltages for indicator operation noted herein are alternating current root-mean-square values. The voltmeter used to measure alternating voltages shall be of a type which is free from frequency errors.

4.4.6 Testing. Unless otherwise specified, the following tests shall be conducted on two indicators operating from one tachometer generator:

a. Room temperature starting

b. Acceleration

c. Scale error at room temperature

d. Low temperature starting

e. Low temperature warm-up

f. Scale error at -54°C

g. Scale error at -35°C

h. Scale error at -5°C

i. Scale error at high temperature

4.5 Test methods.

4.5.1 Examination of product. Each indicator shall be

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inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, proper operation, and performance.

4.5.2 Room temperature starting. This test shall be conducted using the following voltage supplies with frequencies increased from zero cycles per minute up to and not exceeding 150 cpm. The indicator reading taken while the indicator is gently tapped or vibrated shall not be less than 2.5 percent rpm.

4.5.2.1 Variable frequency power supply. One indicator shall be connected to the variable power supply and subjected to the room temperature starting test. The power supply voltage shall be 0.0035 times the frequency in cycles per minute.

4.5.2.2 Tachometer generator. Two indicators shall be connected to a tachometer generator conforming to MIL-G-9398 or MIL-G-26611 and subjected to the room temperature starting test.

4.5.3 Indicator acceleration. The indicators shall be connected to a tachometer generator conforming to MIL-G-9398 or MIL-G-26611 as shown in figure 2. The generator speed shall be increased from zero rpm to 1260 rpm ±210 rpm within a period of 4 seconds. The indicators shall synchronize with the tachometer generator and their indication shall not exceed the room temperature scale error tolerance specified in table I.

4.5.4 Scale error at room temperature. The indicators shall be connected as shown on figure 2 to a generator conforming to MIL-G-9398 or MIL-G-26611 and tested at points of the scale indicated in either table I or table II. The test shall be made either by driving the generator at a speed to produce the specified indicator readings and reading the speed of the generator or by driving the generator at the specified speeds and reading the indicator. The test shall be made with speeds increasing then with speeds decreasing. With speeds increasing, the speed shall be brought up to but shall not exceed the speed specified to give the desired reading, and with speeds decreasing, the speed shall be brought down to but shall not fall below the speed specified to give the desired reading.

4.5.4.1 Test point tolerances. The errors at the test points shall not exceed the tolerances specified in the applicable table. When the speed is held constant at any point of the scale, the pointer shall not oscillate over a range greater than 0.5 percent from 0 to 15 percent and 0.3 percent from 15 to 100 percent.

4.5.5 Friction error. The indicator shall be set at 5, 20, 40, 70, 85, and 100 percent graduations, or at generator speeds 200,

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800, 1600, 3200, 3600, and 4200 rpm whichever the contractor finds more convenient. After each setting, the indicator shall be tapped and the change in indications noted. This change shall not exceed 1.5 percent at the 5 percent graduation, 0.7 percent at the 20 percent graduation, and 0.5 percent at the 40, 70, 85, and 100 percent graduations.

4.5.6 Position error. The indicator shall be operated at 100 percent with the indicator in the normal (dial vertical) position. The indicator reading shall be noted. The maximum deviation from the normal position reading, when tested in the following positions, shall not exceed 0.3 percent:

a. Rotated from normal position through 45° to the rear (dial tilted 45° back from vertical position).

b. Rotated from normal position through 45° to the front (dial tilted 45° forward from vertical position).

c. Rotated from normal position through 45° to the right (dial vertical).

d. Rotated from normal position through 45° to the left (dial vertical).

4.5.7 Pointer alignment. The indicator shall be operated so the main pointer will indicate 90 and 96 percent rpm respectively. The subdial pointer shall then respondingly indicate 0 to 6 respectively within 0.20 percent.

4.5.8 Sealing. The instrument case shall be tested for leaks by means of a mass spectrometer type of helium leak detector. Where a nitrogen-helium mixture is utilized as the filling medium the detected leak rate shall not exceed 0.01 micron cubic foot per hour at a pressure differential of 1 atmosphere. Where 100 percent helium is utilized as the filling medium the leak rate shall not exceed 0.1 micron cubic foot per hour.

*4.5.9 Precipitation or fog. The indicator shall be placed in a chamber, the air temperature of which is controllable. The temperature of this circulating air shall be 85° to 90°C and the indicator shall be kept in this chamber for a minimum of 2 hours. The indicator shall then be removed from the high temperature chamber and within 10 seconds the indicator face shall be immersed to a depth of ¼ inch in a fluid bath maintained at 0° to 5°C. After 5 minutes, the indicator shall be removed from the fluid bath and immediately wiped dry (compressed air shall not be used). The inside of the cover glass and dial face shall be inspected immediately and any evidence of fogging or damage shall be cause for rejection.

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4.5.10 Low temperature starting. Prior to this test, the room temperature scale error test shall be repeated. The indicators shall then be placed in a chamber, the temperature of which is controllable. The temperature of the circulating air shall be reduced to -54° ± 2°C and held at this temperature for the duration of the test. The indicators shall be kept in the chamber for at least 4 hours after the temperature of -54° ± 2°C is reached before the test is started. No voltage shall be applied to the indicator during this period. The indicators, while still at low temperature, shall be subjected to and meet the requirements of 4.5.2.

4.5.11 Low temperature warmup. Following the starting test at low temperature specified in 4.4.10, the indicators shall be maintained at the -54°C temperature for a period of 2 hours or more without voltage applied. The indicators shall then be instantaneously connected to a tachometer generator operating at 4200 rpm. The indicators shall read 100 ± 1.5 percent rpm within a period of 15 seconds.

4.5.12 Scale error at -54°C. Following the low temperature warmup test and while still at -54°C, the indicators readings at the points indicated by asterisks in tables I or II shall be determined in the same manner as specified in 4.5.3 with one generator operating two indicators. The errors at the test points shall not exceed the tolerances specified in the applicable table.

4.5.13 Scale error at -35°C. The indicators readings shall be determined as specified in the scale error test at -54°C except that the temperature of the air in the chamber shall be -35° ±2°C. The indicators shall be kept at -35°C ± 2°C for at least 2 hours prior to the test. The errors at the test points shall not exceed the tolerances specified in the applicable table.

4.5.14 Scale error at -5°C. The indicators readings shall be determined as specified in the scale error test at -54°C, except that the temperature of the air in the chamber shall be -5°C. The indicators shall be kept at -5°C ± 2°C for at least 2 hours prior to the test. The errors at the test points shall not exceed the tolerances specified in the applicable table.

4.5.15 Scale error at high temperature. The indicators scale readings shall be determined as specified in the indicator scale error test at -54°C, except that the temperature of the air in the chamber shall be +71° ± 2°C. The indicators shall be kept in the chamber for 4 hours after the temperature of +71°C is reached before the test is started. The errors at the test points shall not exceed the tolerances specified in the applicable table.

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4.5.16 Insulation. To test the insulation, a potential at a commercial frequency shall be applied between pins A, B, or C of the electrical receptacle and any part of the instrument case or the electrical receptacle for a minimum period of 5 seconds. For the unsealed instrument, the applied voltage shall be 500V. If the indicator is filled with gas and hermetically sealed, the applied voltages shall be 200V for indicators using the helium filling medium and 400V for the nitrogen-helium mixture filled indicators. There shall be no breakdown of the insulation.

4.5.17 Magnetic effect. The indicator, first not operating and second operating at an indication of approximately 90 percent rpm, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5-½ inches from magnetically east or west of the center of the compass and starting directly under the compass, the indicator shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial positions. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in its normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 3°.

4.5.18 Vibration error. The indicator reading shall be determined when the generator is operated at a point between 60 and 90 percent rpm as selected by the inspector. The indicator shall then be subjected to circular vibration as specified under test conditions at an amplitude of 0.009 to 0.011 inch diameter. The indicator readings shall be determined with the generator operating at the original speed as the vibration frequency is varied from 5 to 50 cps. The readings with vibration shall not vary more than 0.7 percent from the readings without vibration. The amplitude of the pointer oscillation shall not exceed 0.7 percent.

4.5.19 Indicator current. The indicator shall be tested using a 21V, 4200 cycle per minute supply to determine that the current drawn by any one of the 3-phase windings does not exceed 0.30 amp.

4.5.20 Low temperature exposure. The room temperature scale error test for increasing speeds only shall be repeated. The indicator shall then be subjected to a temperature of -62° ± 2°C for a period of 48 hours. After the chamber has returned to room temperature and the indicator has remained at room temperature for a minimum period of 4 hours, it shall be subjected to and meet the requirements of the scale error test at room

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temperature, increasing speeds only, and the friction test. There shall be no damage which would adversely affect subsequent operation.

4.5.21 High temperature exposure. The indicator shall be placed within a chamber and the internal temperature of the chamber raised to +71° ± 2°C with an internal relative humidity of not more than 5 percent. The source of heat for the chamber shall be arranged in such a manner that radiant heat shall not fall upon the test specimen. The total volume occupied by a single item of equipment or by several items shall not exceed 50 percent of the internal volume of the test chamber. The indicator shall be maintained at +71° ± 2°C for a period of 24 hours. While still at this temperature, the indicator shall be operated throughout its range, for increasing speeds only, without exhibiting sticking or erratic characteristics. After the chamber has returned to room temperature and the indicator has remained at room temperature for a minimum period of 4 hours, it shall be subjected to and meet the requirements of the scale error at room temperature test, increasing speeds only, and the friction test. There shall be no damage which would adversely affect subsequent operation.

4.5.22 Vibration failure. The indicator shall be mounted on the vibration stand in its normal operating position. The test shall be conducted while the indicator is being operated at an indication of approximately 50 percent. The indicator shall be subjected to vibration with a circle diameter between 0.018 and 0.020 inch for a 3-hour period. The frequency shall be varied uniformly from 5 to 50 cps and return once each hour during this test. After the 3-hour period, the indicator shall be subjected to the scale error at room temperature test, increasing speeds only. The readings shall not differ from the readings before vibration by more than 0.5 percent. In addition, no looseness in the mechanism or damage to any part of the instrument shall result from this test.

4.5.23 Overspeed and reverse operation. One or two of the indicator mechanisms shall be electrically connected to a tachometer generator. The reading of the indicators shall be determined when operated at 20 and 90 percent rpm. The indicators shall be at room temperature and the readings shall be taken before the indicators have attained an appreciable temperature rise.

4.5.23.1 Operation of generator and indicators. The generator and indicators shall then be operated, first at a generator shaft speed of approximately 5,000 rpm for a period of 5

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minutes, and then in the reverse direction of rotation at a generator drive shaft speed of approximately 2,500 rpm for a period of 1 minute.

4.5.23.2 Allowing indicators to cool. After the indicators have been allowed to cool to room temperature, the readings of the indicator shall again be determined when operated at 20 and 90 percent rpm. The change in indicated speed at the respective scale points before and after the overspeed test shall not be more than 0.5 percent.

*4.5.24 Indicator endurance. The readings of the indicator shall be determined when operated at 20 and 90 percent rpm. The indicator shall be at room temperature and the readings shall be taken before the indicator has attained an appreciable temperature rise. The indicator shall then be operated at an indication of 90 ± 5 percent rpm for a period of 1,000 hours. The voltage shall be measured at the indicator terminals and shall be 18.5 ± 1V during the 1,000 hour period. At the end of the 1,000 hour period and after the indicator has been allowed to cool to room temperature, the readings of the indicator shall again be determined when operated at 20 and 90 percent rpm. The change in reading obtained before and after this test shall not be more than 0.5 percent. No screws or other parts shall become loose or damaged as a result of this test. Indicators which have been subjected to all the tests of this specification up to and including this test shall also be subjected to the scale error at room temperature, friction and position error tests. Scale error shall not be 0.5 percent more than allowed at room temperature in tables I and II. Friction error shall not exceed 1.5 percent at the 5 and 20 percent graduations and 1.0 percent at the 40, 70, 85, and 100 percent graduations. Position error shall not exceed 0.5 percent.

4.5.25 Thermal shock. The indicator shall be subjected to four cycles of exposure to water at +85° ± 2°C and +5° ± 2°C without evidence of moisture penetration or damage to casting or enclosures. Each cycle of the test shall consist of immersing the indicator in water at 85°C for a period of 30 minutes and then within 5 seconds of removal from the bath, the indicator shall be immersed for a period of 30 minutes in the other bath maintained at 5°C. This cycle shall be repeated continously; one cycle following the other, until four cycles have been completed. Following this test, the indicator shall be subjected to the sealing test. No indicator leakage shall occur as a result of this test.

4.5.26 Salt spray. The indicator shall be subjected to a salt spray test in accordance with procedure I MIL-E-5272 for a

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period of 50 hours. External connections shall be made to the indicator in such a manner as to simulate installed conditions. Following this test, the indicator shall be examined and there shall be no evidence of external corrosion or deterioration which will affect subsequent operation. The indicator shall then be subjected to the insulation and leak tests as specified herein. The external surface of the indicator may be washed with distilled water and air dried prior to conduction of the insulations test.

*4.5.27 Response. The test will be conducted using two tachometer generators conforming to MIL-G-9398 or MIL-G-26611. One generator will be operating and stabilized at 50 ± 1.5 percent and the other generator will be operating and stabilized at 100 ± 1.5 percent. The indicator to be tested and the generators will be wired using a switch. This will enable the indicator to be driven by either generator. The indicator to be tested will be operated until it reaches operating temperature (90 percent minimum RPM for 5 minutes). The indicator will then be driven a minimum of 1 minute with the slow (50 ± 1.5 percent) speed generator and allowed to stabilize. The switch will then be thrown thereby switching the indicator to the high (100 ± 1.5 percent) speed generator and starting a timer at the same time. The timer will be stopped when the indicator reaches an indication of 100 ± 1.5 percent. The time shown on the timer shall not exceed 2 seconds.

*4.5.28 Internal examination. The indicator shall be examined internally for evidence or corrosion or deterioration resulting from the tests specified herein. There shall be no evidence of corrosion or deterioration.

*4.5.29 Refelctance. The cover glass shall meet the requirements specified in 3.8.2.

*4.6 Inspection and test. Tests of methods of Preservation and Packaging shall be accomplished in accordance with Section 4 of MIL-P-116, to insure compliance with Section 5 of this specification.

5. PACKAGING

*5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified.

*5.1.1 Level A.

*5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 of MIL-P-116.

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*5.1.1.2 Drying. Indicator shall be dried in accordance with process D-4 of MIL-P-116.

*5.1.1.3 Preservation application. Not applicable.

*5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each Indicator, Electrical Tachometer, ERU 6/A, shall be packaged in quantity unit packs of one each in accordance with Method 1C-1 of MIL-P-116. Overbox in PPP-B-636 carton large enough to allow for application of sufficient cushioning material between container and bag, of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the item, when completed packs are subjected to the rough handling drop tests of MIL-P-116.

*5.1.2 Level C. Same as level A except Method III of MIL-P-116 applies in lieu of specified method.

*5.2 Packing. Packing shall be level A, B, or C as specified.

*5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, weather-resistant class, unless otherwise specified by the procuring activity. Insofar as practical, exterior shipping container shall be of uniform shape, size of minimum tare and cube consistent with the protection required. Table II of MIL-STD-794 applies for Army use.

*5.2.2 Level B. Indicator packaged as specified in 5.1.1 shall be packed in domestic class exterior shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 above shall apply.

*5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

6. NOTES

*6.1 Intended use. The tachometer indicator covered by this specification is intended for use to indicate the rotational speed of a jet or prop-jet engine in percent rpm. The calibration is based on the generator's turning 4200 prm at 100 percent rated engine speed. The indicator is to be used where white integral lighting is required. The ERU 6/A indicator is intended for use in aircraft systems that do not require the

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high-efficiency antireflective coating on the cover glass and the ERU 6A/A indicator is intended for use in aircraft systems that do require a high-efficiency antireflective coating on the cover glass.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Level of packaging and packing desired.

c. Whether sampling plan B is to be omitted.

d. Type of indicator required (see 1.2).

6.3 Definitions.

6.3.1 Tapping. The phrase GENTLY TAPPED is defined as meaning the application of a light vibration sufficient to overcome that amount of residual friction normally associated with good quality aircraft instrument.

6.3.2 Hermetic seal. A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic materials, or both. A hermetic seal is not intended to include seals accomplished by gaskets.

6.3.3 Amplitude. Whenever the work AMPLITUDE is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

*6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Hq AFLC CASO/LODS, Federal Center, Battle Creek MI 49016, and information pertaining to qualification of products may be obtained from that activity.

*6.5 International standardization agreement. Certain provisions of this specification are the subject of

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International Standardization Agreement ABC 10/36. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

6.6 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

*6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 99

Preparing Activity:
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